

Narcissistic number(PPDI)

Narcissistic number (自幂数) has another name called Armstrong number (阿姆斯壯数)

1 1-digit Narcissistic number[独身数]

All number 0-9 is Narcissistic number.

2 2-digit Narcissistic numbers

2-digit Narcissistic numbers does not exist.

3 3-digit Narcissistic number(Daffodil number)[水仙花数]

This refers to a 3-digit number where the sum of the cube of its digits equals the number itself.

Example: 153

$$1^3 + 5^3 + 3^3 = 153$$

So 153 is a Daffodil number.

There have 4 daffodil number are:

1. 153
2. 370
3. 371
4. 407

3.1 Find daffodil number with programming

3.1.1 In C++

```

1      #include<iostream>
2      #include<cstdio>
3      int main()
4      {
5          int a, b, c;
6          for (int num = 100; num <= 999; num++) {
7              a = num / 100;
8              b = (num / 10) % 10;
9              c = num % 10;
10             if (a * a * a + b * b * b + c * c * c == num)
11                 printf("%d\n", num);
12         }
13     }

```

3.1.2 In Python

```

1      for num in range(100,1000):
2          a = num // 100
3          b = (num // 10) % 10
4          c = num % 10
5          if a**3+b**3+c**3==num:
6              print (num)

```

4 4-digit Narcissistic number(Four-leaf-Rose Number)[四叶玫瑰数]

This refers to a 4-digit number where the sum of the fourth powers of its digits equals the number itself.

Example: 1634

$$1^4 + 6^4 + 3^4 + 4^4 = 1634$$

So 1634 is a Four-leaf-rose Number.

There have 3 rose number are:

1. 1634
2. 8208
3. 9474

4.1 Find Four-leaf-rose Number with programming.

4.1.1 In C++

```
1      #include<iostream>
2      #include<cstdio>
3      using namespace std;
4
5      int main(){
6          int a,b,c,d;
7          for (int num=1000;num<=9999;num++){
8              a = num / 1000;
9              b = (num / 100) % 10;
10             c = (num / 10) % 10;
11             d = num % 10;
12             if (a*a*a*a+b*b*b*b+c*c*c*c+d*d*d*d==num)
13                 printf("%d\n",num);
14         }
15         cin.get();
16         return 0;
17     }
```

4.1.2 In Python

```
1      for num in range(1000,10000):
2          a = num // 1000
3          b = (num // 100) % 10
4          c = (num // 10) % 10
5          d = num % 10
6          if a**4+b**4+c**4+d**4==num:
7              print (num)
```

5 5-digit Narcissistic number(Five-point-star number)[五角星数]

This refers to a 5-digit number where the sum of the fifth powers of its digits equals the number itself.

There has 3 five-point-star number are:

1. 54748

2. 92727

3. 93084

5.1 Find five-point-star number with programming

```
1      #include<iostream>
2      #include<cstdio>
3      using namespace std;
4      int main(){
5          int a, b, c, d, e;
6          for (int num = 10000; num <= 99999; num++) {
7              a = num / 10000;
8              b = (num / 1000) % 10;
9              c = (num / 100) % 10;
10             d = (num / 10) % 10;
11             e = num % 10;
12             if (a * a * a * a + b * b * b * b + c * c * c
13                 * c + d * d * d * d == num)
14                 printf("%d\n", num);
15         }
16         cin.get()
17         return 0;
18     }
```

```
1      for num in range(10000,100000):
2          a = num // 10000
3          b = (num // 1000) % 10
4          c = (num // 100) % 10
5          d = (num // 10) % 10
6          e = num % 10
7          if a**5+b**5+c**5+d**5+e**5==num:
8              print (num)
```